

Species table and depth-sensing camera/phone table

A compact reference document with two tables: the 13 species in the SPREAD dataset (grouped by plantation and rainforest domain) with their physical characteristics, and a list of cameras and phones with a depth-sensing function.

1. SPREAD dataset species

1.1 Plantation domain (7 species, orchard fruit trees)

Dataset name	Scientific name	Habit	Mature height	Trunk diameter (DBH)	Physical notes
Apple	<i>Malus domestica</i>	Deciduous tree	3–6 m (orchard); up to 12 m unmanaged	~15–30 cm	Rounded, moderately dense crown; simple oval serrated leaves
Lemon	<i>Citrus limon</i>	Evergreen tree	3–6 m	~10–20 cm	Dense, rounded, thorny twigs; glossy elliptic leaves
Loquat	<i>Eriobotrya japonica</i>	Evergreen tree	5–10 m	~20–40 cm	Dense, rounded crown; large, leathery, corrugated leaves
Mango	<i>Mangifera indica</i>	Evergreen tree	10–15 m (orchard); wild/old specimens 30–40 m	Orchard ~30–60 cm; old wild trees can exceed 100 cm	Dense, broad, dome-shaped canopy; long lanceolate leaves
Orange	<i>Citrus sinensis</i>	Evergreen tree	3–10 m (orchard usually kept 4–6 m)	~15–30 cm	Dense rounded evergreen canopy; glossy ovate leaves
Persimmon	<i>Diospyros kaki</i>	Deciduous tree	4–12 m	~20–40 cm	Broad-spreading, moderately dense crown; large glossy leaves
Pomegranate	<i>Punica granatum</i>	Shrub / small tree, often multi-stemmed	2–5 m	Individual stems thin, ~5–15 cm	Bushy, semi-open crown; narrow glossy leaves, orange-red flowers. Single-trunk DBH is not very meaningful given the multi-stem habit

1.2 Rainforest domain (6 species)

Dataset name	Scientific name	Habit	Mature height	Trunk diameter (DBH)	Physical notes
AliiFig	<i>Ficus binnendijkii</i> 'Alii' (best-fit guess, high confidence)	Tree / large shrub	3–9 m (cultivated)	~10–25 cm, often multi-stemmed	Long narrow glossy "banana-leaf" foliage, dense weeping habit
BangaloPalm	<i>Archontophoenix cunninghamiana</i> (best-fit guess, high confidence)	Palm	10–20 m	~20–30 cm, swollen at the base	Feather (pinnate) fronds, crown spread 3–5 m
Fern	<i>Genus unresolved</i>	Fern	Ground fern <1–2 m with no true trunk, or tree fern (e.g. Cyathea) 2–10 m	Ground fern: not meaningful. Tree fern: ~15–30 cm	Dense frond clump; excluded from DBH supervision in the dataset
LeechVine	<i>Unidentified</i> (low confidence)	Liana / climbing vine	Unknown	Unknown	Irregular shape; lowest detection F1 among the rainforest classes; excluded from DBH supervision in the dataset

Dataset name	Scientific name	Habit	Mature height	Trunk diameter (DBH)	Physical notes
RubberFig	<i>Ficus elastica</i> (best-fit guess, high confidence)	Tree (banyan-group fig)	10–15 m (cultivated); wild specimens 30–40 m	Cultivated ~20–50 cm; old wild trees >100–200 cm	Large, thick, glossy leaves; aerial roots on mature trees
Umbrella	<i>Schefflera arboricola</i> (best-fit guess, medium confidence)	Shrub / small tree	2–4 m (cultivated); up to 9 m in native habitat	Generally thin, <10–15 cm	Palmate compound leaves radiating like umbrella spokes; dense rounded crown

Scientific names for AliiFig, BangaloPalm, RubberFig, and Umbrella are best-fit identifications from matching the common asset name to Quixel Megascans plant assets used in the Unreal Engine 5 renders, not official labels from the SPREAD paper. LeechVine could not be matched to any real species.

2. Cameras and phones with a depth-sensing function

2.1 RGB-D cameras

Name	Technology	Specifications	Year	Price (USD)
Microsoft Kinect (v1, for Xbox 360)	Structured light (IR laser projector + monochrome CMOS IR sensor, PrimeSense-based)	Depth range ~1.2-3.5 m (near mode ~0.4-3m); depth resolution 640x480 (VGA), 11-bit / 2048 levels; RGB 640x480; FOV 57deg horizontal x 43deg vertical; depth resolution ~0.75mm/px at 50cm, ~3mm/px at 2m	2010	\$149.99 at launch (bundled/standalone Xbox 360 peripheral)
Microsoft Kinect v2 (for Xbox One)	Time-of-Flight (ToF)	Depth range ~0.4-4.5 m (mm-level precision); depth resolution 512x424 @ 30fps; RGB 1920x1080; IR stream 512x424; FOV ~70deg x 60deg (also cited as 84.1deg x 53.8deg H/V)	2013	~\$149-199 (standalone 'Kinect for Windows v2' sensor); bundled originally with \$499 Xbox One
Orbbec Astra (original series)	Structured light (IR pattern projector + IR camera, PrimeSense-alternative ASIC)	Depth range ~0.4-8 m depending on variant (Astra Mini S: 0.4-2m; Astra/Astra Pro: 0.6-5m); depth & RGB resolution up to 640x480 @ 30fps (VGA); single USB cable for power+data	2015	~\$149-199 at launch (Astra Pro Indiegogo early-bird price was \$99)
Intel RealSense Depth Camera D415	Active/passive stereo IR (IR stereo pair + IR pattern projector, rolling shutter)	Depth range ~0.5-3 m (ideal); depth resolution up to 1280x720 @ 30fps (up to 1280x720/90fps in some modes); RGB 1920x1080 @ 30fps; depth FOV ~65deg x 40deg (H69.4 x V42.5 RGB FOV); USB 3.1 Gen1 Type-C	2018	~\$149 at launch (2018); ~\$270 current list price
Intel RealSense Depth Camera D435i	Active stereo IR (global shutter stereo pair + IR projector) plus onboard IMU	Depth range ~0.3-3 m ideal (0.2-10 m operating); depth resolution up to 1280x720 @ 90fps; RGB 1920x1080 @ 30fps; FOV ~87deg x 58deg; BMI055 6-DoF IMU; USB 3.1 Type-C	2019	~\$199 at launch (2019); ~\$330 current list price
Occipital Structure Sensor (Pro)	Structured light (IR pattern projector), mounts to iPad/iPhone for mobile 3D scanning	Working range ~0.4-3.5 m; accuracy ~+/-0.29% (Pro/healthcare-grade version); mobile/handheld form factor, Lightning/USB-C bracket mount to iOS devices	2019	\$399-599 depending on bundle
Intel RealSense Depth Camera D455	Active stereo IR (global shutter, wider 95mm baseline) plus IMU	Depth range ~0.6-6 m; depth resolution up to 1280x720 @ 90fps; RGB 1920x1080 (global shutter); FOV ~86deg x 57deg; depth error <2% at 4m; IMU for motion refinement; USB 3.1	2020	~\$239 at launch (2020)
Microsoft Azure Kinect DK	Time-of-Flight (ToF) - 1MP AMCW (amplitude-modulated continuous-wave) iToF depth sensor	Depth range up to 5.46 m (mode-dependent: NFOV ~0.5-3.86m, WFOV ~0.25-2.88m or 2.21-5.46m); depth resolution up to 1024x1024 (NFOV) or 512x512 (WFOV); RGB 12MP; 7-mic circular array; IMU; USB-C. Discontinued Oct 2023, SDK/firmware support through 2027.	2020	\$399 at launch
Stereolabs ZED 2i	Passive stereo (dual RGB global-shutter/rolling-shutter pair) with neural-network depth estimation (Neural Depth Engine)	Depth range up to 20 m (sub-mm accuracy at close range); resolution up to 4416x1242 @ 15fps or 1920x1080 @ 60fps; FOV 110deg (H) x 70deg (V) x 120deg (D); IP66-rated enclosure; 9-DoF IMU; USB 3.0 Type-C	2021	~\$549-749 depending on lens/IP configuration (exact launch price not publicly confirmed)
Luxonis OAK-D Pro / OAK-D S2	Active stereo IR (stereo pair + IR dot projector + IR illumination LED for low light) with onboard AI (Myriad X VPU)	Ideal depth range ~0.7-12 m (Pro); 12MP autofocus/fixed-focus central RGB camera option; 940nm notch-filtered stereo pair; 4 TOPS onboard neural inference; USB3/PoE variants available	2021	OAK-D S2 ~\$329; OAK-D Pro ~\$399 (varies by variant: PoE, wide-FOV, etc.)

Name	Technology	Specifications	Year	Price (USD)
Intel RealSense Depth Camera D457	Active stereo IR plus IMU, industrial GMSL2/FAKRA interface (vs USB on other D400 models)	Depth range ~0.6-6 m; depth resolution up to 1280x720 @ 90fps; dimensions 124 x 29 x 36 mm; IP65-rated enclosure; GMSL/FAKRA coax connection up to 15 m cable run, supports multi-camera sync over one cable	2022	~\$520-530
Orbbec Femto Bolt	Time-of-Flight (ToF) - 1MP iToF sensor, positioned as an Azure Kinect DK replacement (uses Microsoft ToF tech)	Depth range ~0.25-5.46 m (mode dependent); depth resolution up to 1024x1024 @ 15fps or 640x576 @ 30fps; RGB 4K, 90deg FOV; depth FOV 120deg; std deviation of random error <=17mm; USB-C for power+data	2023	Not publicly listed by Orbbec (comparable positioning/tier to the discontinued \$399 Azure Kinect DK)
Stereolabs ZED X	Passive stereo (dual 2.3MP global-shutter sensors) with neural depth estimation, GMSL2 industrial interface	Depth range 0.3-20 m (2.2mm accuracy) or 1-35 m (4mm accuracy); resolution 1200p @ 60fps global shutter; IP67-rated aluminum enclosure; vibration-resistant IMU (16-bit accel+gyro); GMSL2 low-latency interface	2023	€599 (~\$650)

2.2 Phones and tablets

Name	Technology	Specifications	Year	Price (USD)
Oppo R17 Pro / RX17 Pro	Time-of-Flight (ToF) - 3D stereo ToF rear camera	Triple rear camera: 12MP main (variable aperture) + 20MP secondary + a 3D TOF stereo camera module for AR; among the first Chinese-brand phones with a ToF depth camera (2018)	2018	€599 (RX17 Pro, European launch)
Samsung Galaxy S10 5G	Time-of-Flight (ToF) - 'DepthVision' 3D depth sensing camera	Quad rear camera including a VGA-class ToF depth sensor for AR mapping and portrait background blur; 6.7in display	2019	~\$1,300 (Verizon exclusive at launch)
Samsung Galaxy Note10+	Time-of-Flight (ToF) - 'DepthVision' VGA depth camera	Fourth rear camera is a VGA ToF DepthVision sensor enabling 3D scanning/AR measurement and live focus; note the base Note10 (non-Plus) omits the ToF sensor	2019	\$1,099.99 at launch
Huawei P30 Pro	Time-of-Flight (ToF) rear camera	Quad rear camera including a dedicated ToF sensor for AR depth/scale detection and portrait effects; also known for 5x optical / 10x hybrid zoom	2019	€999 (128GB) at launch
Huawei Mate 30 Pro	Time-of-Flight (ToF) - dual ToF sensors (front and rear)	Quad rear camera with a 3D ToF sensor plus a second 3D ToF sensor on the front camera array, used for depth-based AR, gesture control, and portrait effects	2019	Not publicly confirmed at global launch (regional pricing only; Huawei's US market exit limited official listings)
Honor View20 (V20)	Time-of-Flight (ToF) rear 3D camera	Dual rear camera: 48MP main (Sony IMX586) plus a 3D ToF sensor used for object/body scanning and AR effects; 6.4in display	2019	~\$480 (CNY 3,299, 6+128GB China price) / £499 (UK base model)
LG G8 ThinQ	Time-of-Flight (ToF) - front-facing 'Z Camera'	Front-facing Z Camera ToF sensor used for 'Hand ID' palm-vein authentication and 'Air Motion' touchless gesture control; rear dual camera is conventional (no ToF)	2019	\$819-900 at launch
Apple iPhone 12 Pro / 13 Pro (and Pro Max variants) - LiDAR	LiDAR (direct Time-of-Flight scanning laser)	LiDAR Scanner range up to ~5 m; used for instant AR plane detection, Night mode portraits, and ARKit depth APIs; iPhone 12 Pro released with 6.1in display, A14 Bionic	2020	\$999 at launch (128GB, iPhone 12 Pro); iPhone 13 Pro also launched at \$999 in 2021

Name	Technology	Specifications	Year	Price (USD)
Apple iPad Pro (2020, 4th generation, and later LiDAR models)	LiDAR (direct Time-of-Flight scanning laser)	LiDAR Scanner range up to ~5 m; paired with wide + ultra-wide rear cameras; targeted at AR and 3D scanning/measuring apps	2020	\$799 (11-inch) / \$999 (12.9-inch) at launch
Samsung Galaxy S20+ / S20 Ultra	Time-of-Flight (ToF) - 'DepthVision' 0.3MP depth sensor	Rear quad/penta camera array including a 0.3MP ToF DepthVision sensor for bokeh simulation and AR; note: Samsung dropped the ToF sensor starting with the Galaxy S21 generation, and the S23 Ultra has only a laser-AF rangefinder (not a full ToF depth camera) - verified, so S23 Ultra is excluded from this catalog	2020	\$1,399-\$1,400 at launch (S20 Ultra)
Apple iPhone 14 Pro / 15 Pro / 16 Pro (and Pro Max variants) - LiDAR	LiDAR (direct Time-of-Flight scanning laser), same core sensor lineage as iPhone 12/13 Pro with software/ISP refinements	LiDAR Scanner range up to ~5 m; supports ARKit scene depth/mesh APIs, Night mode autofocus, Cinematic mode depth, spatial video (iPhone 15/16 Pro)	2022	\$999+ at launch across generations (iPhone 14/15/16 Pro)

The Galaxy S23 Ultra is not included since it only has a laser-AF rangefinder rather than a full ToF depth sensor; Samsung dropped ToF after the S20 generation. Product photos and image source or license notes are in the separate catalog document (*Katalog_Kamera_Handphone_Depth_Sensing.pdf*).